arXiv:2302.00532v1 [math.AP] 1 Feb 2023

SOME ESTIMATES FOR MITTAG-LEFFLER FUNCTION IN QUANTUM CALCULUS AND APPLICATIONS

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Abstract. The study of the Mittag-Leffler function and its various generalizations has become a very popular topic in mathematics and its applications. In the present paper we prove the following estimate for the q-Mittag-Leffler function:

$$\frac{1}{1+\Gamma_q(1-\alpha)z} \leq e_{\alpha,1}(-z;q) \leq \frac{1}{1+\Gamma_q(\alpha+1)^{-1}z}.$$

for all $0<\alpha<1$ and $z>0$.

Moreover, we apply it to investigate the solvability results for direct and inverse problems for time-fractional pseudo-parabolic equations in quantum calculus for a large class of positive operators with discrete spectrum.

1. Introduction

The theory of quantum calculus analyses the basic structures of physics from the point of view of non-commutative geometry on quantum groups. We can obtain a quantum group by considering the classical algebra of observables and quantising it relative to some Poisson bracket. It is well-known that quantum algebras and quantum groups support the theory of several physical phenomena. They play an important role in the conformal field theory, in statistical physics [5] and in a wide range of applications, from cosmic strings and black holes to solid state physics problems [4], [8], [22] and [23] and [44].

In [7] and [25], L. Biedenharn and A. Macfarlane showed that the q-calculus plays a central role in the representation of the quantum groups with a deep physical meaning and is not merely a mathematical exercise. In the framework of the q-Heisenberg algebra, q-deformed Schrödinger equations have been proposed in [41] and [42]. A number of articles address the further developments and recent results in the q-deformed calculus (see e.g. [3], [6], [10], [26], [37], [38] and we refer to the books [11] and [13]).

Recently, it is also well recognized that the Mittag-Leffler function plays a fundamental role in the fractional calculus problems. In 1903, the Swedish mathematician

2010 *Mathematics Subject Classification.* 34C10, 39A10, 26D15.

Key words and phrases. q-heat equation, q-derivative, q-calculus .

The authors are supported by the FWO Odysseus 1 grant G.0H94.18N: Analysis and Partial Differential Equations and by the Methusalem programme of the Ghent University Special Research Fund (BOF) (Grant number 01M01021). Michael Ruzhansky is also supported by EPSRC grant EP/R003025/2, and the second author by the international internship program "Bolashak" of the Republic of Kazakhstan.

Gösta Mittag-Leffler discovered [28] the following function defined by

$$E_{\alpha}(z)=\sum_{k=0}^{\infty}\frac{z^k}{\Gamma(\alpha k+1)},\quad \Re(\alpha)>0, z\in\mathbb{C},$$

which is convergent in the whole complex plane. In [43] (see [19, Subsection 1.8]), A. Wiman firstly introduced the generalization of the Mittag-Leffler as a two-parameter version $E_{\alpha,\beta}(z)$ defined by

$$E_{\alpha,\beta}(z)=\sum_{k=0}^{\infty}\frac{z^k}{\Gamma(\alpha k+\beta)},$$

respectively, where $z,\beta\in\mathbb{C}$ and $\alpha>0$. The following Mittag-Leffler function estimate is known [39, Theorem 4]:

$$\frac{1}{1+\Gamma(1-\alpha)z}\leq E_{\alpha,1}(-z)\leq\frac{1}{1+\Gamma(\alpha+1)^{-1}z}, \tag{1.1}$$

and, in particular, from this inequality it follows that

$$0<E_{\alpha,1}(-z)<1, \tag{1.2}$$

for $0<\alpha<1$ and $z>0$. Moreover, we get the the estimate in the following form

$$|E_{\alpha,\beta}(-z)|\leq\frac{C}{1+z}, \tag{1.3}$$

where $z>0$, $0<\alpha<2$, $\beta\in\mathbb{R}$ and $C>0$ (see [1] and [28, Subsection 2.1]. These estimates of Mittag-Leffler functions have been widely applied to the solutions of certain problems formulated in terms of fractional order differential, integral and difference equations (see [15], [19], [28] and [40]). Thus, it has recently become a subject of interest for many authors in the field of fractional calculus and its applications on inverse source problems for the diffusion, sub-diffusion and for other types of equations. Motivated by these avenues of applications, there are numerous works published only in recent years in this area (for details, see [2], [33] and [34]).

In [24], Z. S. I. Mansour has introduced a q-analogue of the Mittag-Leffler function, and obtained a fundamental set of solutions for the homogeneous linear sequential q-difference equations with constant coefficients, and a general solution for the corresponding nonhomogeneous equations (see also [3]). The q-Mittag-Leffler function due to Mansour [24], and the q-Mittag-Leffler function $e_{\alpha,\beta}(z;q)$ are given by

$$e_{\alpha,\beta}(z;q)=\sum_{k=0}^{\infty}\frac{z^k}{\Gamma_q(\alpha k+\beta)},\quad |z(1-q)^{\alpha}|<1, \tag{1.4}$$

when $\alpha>0$, $\beta\in\mathbb{C}$, and Γ_q is the q-Gamma function (see (2.9)).

In the present paper, we establish estimates (1.1), (1.2) and (1.3) for the function (1.4) (see Lemmas 3.1 and 3.2). Moreover, we apply these estimates to prove existence and uniqueness of solutions of direct and inverse problems for time-fractional pseudo-parabolic equations with the Caputo fractional operator in the quantum calculus, and a self-adjoint operator in the abstract setting of Hilbert spaces.

This paper is organized as follows: In Section 2, we recall some necessary fundamental concepts of the quantum calculus. In Section 3, we give some lemmas to

prove our main results. In Sections 4 and 5, we apply these results to study the existence and uniqueness of generalized solutions for the direct problems and the inverse problems (see Theorem 4.1, Theorem 4.2 and Theorem 5.1).

2. Preliminaries

For the convenience of the reader, in this section, we recall some necessary concepts and definitions of the quantum calculus, for which we refer to [9, 10, 13] . Moreover, we give lemmas will be used to prove of our main results. Throughout this paper, we assume that $0 < q < 1$.

Let $\alpha \in \mathbb{R}$. Then a q-real number $[\alpha]_q$ is defined by

$$[\alpha]_q := \frac{1-q^\alpha}{1-q},$$

where $\lim\limits_{q\to 1} \frac{1-q^\alpha}{1-q} = \alpha$.

For $a \in \mathbb{R}$ and $n = 0, 1, \cdots$, the q-shifted factorials are defined by

$$(a;q)_0 = 1, \quad (a;q)_n = \prod_{k=0}^{n} \left(1 - q^k a\right), \quad (a;q)_\infty = \lim_{n\to\infty} (a;q)_n. \tag{2.1}$$

Note that

$$(a;q)_n = \frac{(a;q)_\infty}{(aq^n;q)_\infty}, \quad (a;q)_{2n} = (a;q^2)_n (aq;q^2)_n. \tag{2.2}$$

The function $e_q(x)$ is a q-exponential function defined by its q-Taylor series ([9]):

$$e_q(x) = \sum_{k=0}^{\infty} \frac{x^k}{[k]_q!}, \tag{2.3}$$

where

$$[k]_q! = \begin{cases} 1, & \text{if } k = 0, \\ [1]_q \times [2]_q \times \cdots \times [n]_q, & \text{if } k \in \mathbb{N}. \end{cases}$$

Recall that the q-difference operator (or the q-derivative), acting on functions of the variable x, D_q is defined by

$$D_q f(x) = \frac{f(x) - f(qx)}{x(1-q)}. \tag{2.4}$$

Note that if f is differentiable at x, then $\lim\limits_{q\to 1} D_q f(x) = f'(x)$, and the operator D_q is called the Euler–Jackson difference operator (see [14]). Note that

$$D_q \left[x^\alpha (s/x;q)_\alpha\right] = -[\alpha]_q x^{\alpha-1} (qs/x;q)_{\alpha-1}, \quad \alpha \in \mathbb{R}, \tag{2.5}$$

where

$$a^\alpha (s/a;q)_\alpha = a^\alpha \frac{(s/a;q)_\infty}{(q^\alpha s/a;q)_\infty}.$$

The q-translation operator ε^y is presented by M. E. H. Ismail in [16] and is defined by

$$\varepsilon^y x^\alpha = x^\alpha (y/x;q)_\alpha,$$

and its extension is given as follows (see, [3, Section 1.2]):

$$\varepsilon^y\left(\sum_{k=0}^{\infty}\alpha_k x^\alpha\right)=\sum_{k=0}^{\infty}\alpha_k x^\alpha (y/x;q)_\alpha. \tag{2.6}$$

Therefore, using (2.6) we can take a more general form of (1.4) as follows

$$\varepsilon^y e_{\alpha,\beta}\left(z^\gamma;q\right)=\sum_{k=0}^{\infty}\frac{x^{\alpha k}(y/x;q)_{\alpha k}}{\Gamma_q(\alpha k+\beta)}. \tag{2.7}$$

The q-integral (or Jackson integral) is defined as (see [17])

$$\int_0^a f(x)d_qx=(1-q)a\sum_{m=0}^{\infty}q^m f(aq^m)$$

and

$$\int_0^\infty f(x)d_qx=(1-q)\sum_{m=-\infty}^{\infty}q^m f(q^m),$$

provided that the sums converge absolutely. The q-Jackson integral in the interval $[a,b]$ is defined by

$$\int_a^b f(x)d_qx=\int_0^b f(x)d_qx-\int_0^a f(x)d_qx.$$

The formula for the q–integration by parts is

$$\int_a^b u(x)D_qv(x)d_qx=[u(x)v(x)]_a^b-\int_a^b v(qx)D_qu(x)d_qx. \tag{2.8}$$

For any $x>0$, the q–gamma function is defined by

$$\Gamma_q(x)=\frac{(q;q)_\infty}{(q^x;q)_\infty}(1-q)^{1-x},\quad \Gamma_q\left(x+1\right)=\Gamma_q\left(x\right)[x]_q. \tag{2.9}$$

The Riemann-Liouville q-fractional integral $I^\alpha_{q,a+}f$ of order $\alpha>0$ is defined by ([27])

$$\left(I^\alpha_{q,a+}f\right)(x)=\frac{1}{\Gamma_q\left(\alpha\right)}\int_a^x(x-qt)_q^{\alpha-1}f\left(t\right)d_qt.$$

The fractional q-derivative of Caputo type of order $\alpha>0$ is defined by ([27])

$$\left({}^cD^\alpha_{q,a+}f\right)(x)=\left(I^{[\alpha]-\alpha}_{q,a+}D^{[\alpha]}_{q,a+}f\right)(x), \tag{2.10}$$

where $[\alpha]$ denotes the smallest integer greater or equal to α.

Notation. The symbol $M\lesssim K$ means that there exists $\gamma>0$ such that $M\leq\gamma K$, where γ is a constant. If $M\lesssim K\lesssim M$, then we write $M\approx K$.

3. Some estimates of the q-Mittag-Leffler function

In this section we establish estimates (1.1), (1.2) and (1.3) for the function (1.4) in the following lemmas.

Lemma 3.1. *If $0<\alpha<1$ and $z>0$. Then*

$$\frac{1}{1+\Gamma_q(1-\alpha)z}\leq e_{\alpha,1}(-z;q)\leq\frac{1}{1+\Gamma_q(\alpha+1)^{-1}z}. \tag{3.1}$$

In particular, we have

$$0<e_{\alpha,1}(-z;q)<1. \tag{3.2}$$

Proof. The lower estimate. We assume that $0<\alpha<1$. Then using (2.1) we get that

$$\begin{aligned} q/q^{1-\alpha}=q^{\alpha}\leq 1\Rightarrow q\leq q^{1-\alpha} \quad &\Rightarrow \quad 1-q^{1-\alpha}\leq 1-q \\ &\Rightarrow \quad \prod_{n=0}^{\infty}(1-q^nq^{1-\alpha})\leq\prod_{n=0}^{\infty}(1-q^nq) \\ &\overset{(2.1)}{\Rightarrow} \quad \gamma_{1,q}:=\frac{(q;q)_{\infty}}{(q^{1-\alpha};q)_{\infty}}\geq 1; \end{aligned} \tag{3.3}$$

$$\begin{aligned} q^k/q^{k\alpha}=q^{k(1-\alpha)}\leq 1\Rightarrow q^k\leq q^{k\alpha} \quad &\Rightarrow \quad 1-q^{\alpha k+1}\leq 1-q^{k+1} \\ &\overset{(2.1)}{\Rightarrow} \quad \gamma_{2,q}:=\frac{\left(q^{\alpha k+1};q\right)_{\infty}}{(q^{k+1};q)_{\infty}}\leq 1; \end{aligned} \tag{3.4}$$

$$\begin{aligned} q^{k+1}/q^{\alpha k+1-\alpha}=q^{k(1-\alpha)+\alpha}\leq 1 \quad &\Rightarrow \quad q^{k+1}\leq q^{\alpha k+1-\alpha} \\ &\Rightarrow \quad 1-q^{\alpha k+1-\alpha}\leq 1-q^{k+1} \\ &\overset{(2.1)}{\Rightarrow} \quad \gamma_{3,q}:=\frac{\left(q^{\alpha k+1-\alpha};q\right)_{\infty}}{(q^{k+1};q)_{\infty}}\leq 1, \end{aligned} \tag{3.5}$$

for $k=0,1,2,3,\cdots$.

Applying (2.9) and these inequalities (3.3)-(3.5), we find that

$$\begin{aligned} N_q(k) \quad &:= \quad \frac{\left(q^{\alpha k+1};q\right)_{\infty}}{(q;q)_{\infty}}(1-q)^{\alpha k}-\frac{\left(q^{\alpha(k-1)+1};q\right)_{\infty}}{(q^{1-\alpha};q)_{\infty}}(1-q)^{\alpha k} \\ &\overset{(3.3)(3.4)}{=} \quad (1-q)^{\alpha k}\left[\gamma_{2,q}\frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}}-\gamma_{1,q}\frac{\left(q^{\alpha(k-1)+1};q\right)_{\infty}}{(q;q)_{\infty}}\right] \\ &\leq \quad (1-q)^{\alpha k}\left[\frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}}-\frac{\left(q^{\alpha k+1-\alpha};q\right)_{\infty}}{(q;q)_{\infty}}\right] \\ &\overset{(3.5)}{=} \quad \frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}}(1-q)^{\alpha k}\left[1-\gamma_{3,q}\right] \\ &\leq \quad (1-q)^{\alpha k-k}\left[(1-q)^{1-(k+1)}\frac{(q;q)_{\infty}}{(q^{k+1};q)_{\infty}}\right]^{-1} \\ &\overset{(2.9)}{=} \quad \frac{(1-q)^{(\alpha-1)k}}{\Gamma_q(k+1)}. \end{aligned} \tag{3.6}$$

Thus,

$$N_q(k) \lesssim \frac{(1-q)^{(\alpha-1)k}}{\Gamma_q(k+1)}. \tag{3.7}$$

We assume that $0<\epsilon<1-\alpha$. Then using (2.1) we write the following inequalities

$$\begin{aligned} q^{\alpha k+\alpha}/q^{\alpha k+1} = q^{\alpha-1} \geq 1 \Rightarrow q^{\alpha k+1} \leq q^{\alpha k+\alpha} \quad &\Rightarrow \quad 1-q^{\alpha k+\alpha} \leq 1-q^{\alpha k+1} \\ &\overset{(2.1)}{\Rightarrow} \quad \left(q^{\alpha k+\alpha};q\right)_\infty \leq \left(q^{\alpha k+1};q\right)_\infty; \end{aligned} \tag{3.8}$$

$$\begin{aligned} q^{k+1}/q^{\alpha k+1-\alpha} = q^{k(1-\alpha)+\alpha} \leq 1 \quad &\Rightarrow \quad q^{k+1} \leq q^{\alpha k+1-\alpha} \\ &\overset{(2.1)}{\Rightarrow} \quad \left(q^{\alpha k+1-\alpha};q\right)_\infty \leq \left(q^{k+1};q\right)_\infty; \end{aligned} \tag{3.9}$$

$$\begin{aligned} q/q^{1-\alpha} = q^{\alpha} \leq 1 \quad &\Rightarrow \quad q \leq q^{1-\alpha} \\ &\overset{(2.1)}{\Rightarrow} \quad \left(q^{1-\alpha};q\right)_\infty \leq (q;q)_\infty; \end{aligned} \tag{3.10}$$

$$\left.\begin{aligned} q^{1-\alpha}/(q^\epsilon/3) = 3q^{1-\alpha-\epsilon} \leq 3 \Rightarrow q^{1-\alpha} \leq 3(q^\epsilon/3) \quad &\overset{(2.1)}{\Rightarrow} 3 \leq \frac{\left(q^{1-\alpha};q\right)_\infty}{(q^\epsilon/3;q)_\infty} \\ q^{k+1}/q^{\alpha k+\alpha} = q^{-\alpha(k+1)} \geq 1 \Rightarrow q^{\alpha k+\alpha} \leq q^{k+1} \quad &\overset{(2.1)}{\Rightarrow} 1 \leq \frac{\left(q^{\alpha k+\alpha};q\right)_\infty}{\left(q^{k+1};q\right)_\infty} \end{aligned}\right\} \Rightarrow$$

$$\Rightarrow \gamma_{4,q} := \frac{(q^{1-\alpha};q)_\infty}{(q^\epsilon/3;q)_\infty}\frac{\left(q^{\alpha k+\alpha};q\right)_\infty}{\left(q^{k+1};q\right)_\infty} \geq 3. \tag{3.11}$$

Let $q_0 := q^\epsilon/3$ and $\gamma_{5,q} := \frac{1-q_0^{1-\epsilon\alpha}}{q^{\epsilon(1-\epsilon\alpha)}(1-q_0)}$. Then $\lim\limits_{\epsilon\to 0}\gamma_{5,q}=1$ and

$$\begin{aligned} qq_0/q = q_0 \leq 1 \Rightarrow qq_0 \leq q \quad &\Rightarrow \quad 1-q \leq 1-qq_0 \\ &\overset{(2.1)}{\Rightarrow} \quad (q;q)_\infty \leq (qq_0;q)_\infty, \end{aligned} \tag{3.12}$$

and applying above inequalities and (2.1) and (3.8)-(3.12) we have that

$$\begin{aligned} \widehat{N}_q(k) \quad &:= \quad \gamma_{5,q}\frac{\left(q^{\alpha k+1};q\right)_\infty}{(q;q)_\infty} - q^\epsilon\frac{\left(q^{\alpha(k-1)+1};q\right)_\infty}{(q^{1-\alpha};q)_\infty} \\ &\overset{(3.8)(3.9)(3.12)}{\geq} \quad \frac{1-q_0^{1-\epsilon\alpha}}{q^{\epsilon(1-\epsilon\alpha)}(1-q_0)}\frac{\left(q^{\alpha k+\alpha};q\right)_\infty}{(qq_0;q)_\infty} - q^\epsilon\frac{\left(q^{k+1};q\right)_\infty}{(q^{1-\alpha};q)_\infty} \\ &\overset{(2.1)}{=} \quad \frac{1-q_0^{1-\epsilon\alpha}}{q^{\epsilon(1-\epsilon\alpha)}}\frac{\left(q^{\alpha k+\alpha};q\right)_\infty}{(q_0;q)_\infty} - q^\epsilon\frac{\left(q^{k+1};q\right)_\infty}{(q^{1-\alpha};q)_\infty} \\ &= \quad \frac{\left(q^{k+1};q\right)_\infty}{(q^{1-\alpha};q)_\infty}\left[\frac{1-3^{\epsilon\alpha-1}q^{\epsilon(1-\epsilon\alpha)}}{q^{\epsilon(1-\epsilon\alpha)}}\gamma_{4,q} - q^\epsilon\right] \\ &\overset{(3.11)}{\geq} \quad \frac{\left(q^{k+1};q\right)_\infty}{(q^{1-\alpha};q)_\infty}\left[3\frac{(1-3^{\epsilon\alpha-1}q^{\epsilon(1-\epsilon\alpha)})}{q^{\epsilon(1-\epsilon\alpha)}} - q^\epsilon\right] \\ &\overset{(3.10)}{\geq} \quad \frac{\left(q^{k+1};q\right)_\infty}{(q;q)_\infty}3\left[q^{-\epsilon(1-\epsilon\alpha)} - 3^{\epsilon\alpha-1} - q^\epsilon/3\right]. \end{aligned} \tag{3.13}$$

From (2.9), (3.6) and (3.13), we get that

$$\begin{aligned}
N_q(k) &\overset{(3.6)}{=} (1-q)^{\alpha k}\left[\frac{\left(q^{\alpha k+1};q\right)_\infty}{(q;q)_\infty}-\frac{\left(q^{\alpha(k-1)+1};q\right)_\infty}{\left(q^{1-\alpha};q\right)_\infty}\right]\\
&\overset{(3.13)}{=} (1-q)^{\alpha k}\lim_{\epsilon\to 0}\widehat{N}_q(k)\\
&\geq (1-q)^{\alpha k}\frac{\left(q^{k+1};q\right)_\infty}{(q;q)_\infty}3\left[1-1/3-1/3\right]\\
&\geq (1-q)^{\alpha k-k}\left[(1-q)^{1-(k+1)}\frac{(q;q)_\infty}{\left(q^{k+1};q\right)_\infty}\right]^{-1}\\
&\overset{(2.9)}{=} \frac{(1-q)^{(\alpha-1)k}}{\Gamma_q(k+1)},
\end{aligned}$$

for $k=0,1,2,3,\cdots$.

Hence,

$$\frac{(1-q)^{(\alpha-1)k}}{\Gamma_q(k+1)}\lesssim N_q(k). \tag{3.14}$$

By using (1.4), (3.6), (3.7) and (3.14) we find

$$\begin{aligned}
\frac{e_{\alpha,1}(-z;q)}{\left[1+\Gamma_q(1-\alpha)z\right]^{-1}} &= e_{\alpha,1}(-z;q)+e_{\alpha,1}(-z;q)\Gamma_q(1-\alpha)z\\
&\overset{(1.4)}{=} \sum_{k=0}^{\infty}\frac{(-z)^k}{\Gamma_q(\alpha k+1)}+\sum_{k=0}^{\infty}\frac{(-z)^{k+1}\Gamma_q(1-\alpha)}{\Gamma_q(\alpha k+1)}\\
&= 1+\sum_{k=0}^{\infty}\frac{(-z)^k}{\Gamma_q(\alpha k+1)}-\left[1+\sum_{k=1}^{\infty}\frac{(-z)^k\Gamma_q(1-\alpha)}{\Gamma_q(\alpha(k-1)+1)}\right]\\
&= 1+\sum_{k=0}^{\infty}\frac{(-z)^k}{\Gamma_q(\alpha k+1)}-\sum_{k=0}^{\infty}\frac{(-z)^k\Gamma_q(1-\alpha)}{\Gamma_q(\alpha(k-1)+1)}\\
&= 1+\sum_{k=0}^{\infty}(-z)^k\left[\frac{1}{\Gamma_q(\alpha k+1)}-\frac{\Gamma_q(1-\alpha)}{\Gamma_q(\alpha(k-1)+1)}\right]\\
&\overset{(2.1)}{=} 1+\sum_{k=0}^{\infty}(-z)^k\left[\frac{\left(q^{\alpha k+1};q\right)_\infty}{(q;q)_\infty}(1-q)^{\alpha k}\right.\\
&- \left.\frac{\left(q^{\alpha(k-1)+1};q\right)_\infty}{\left(q^{1-\alpha};q\right)_\infty}(1-q)^{\alpha k}\right]
\end{aligned}$$

$$\begin{aligned}
&\overset{(3.6)}{=} && 1+\sum_{k=0}^{\infty}(-z)^k N_q(k)\\
&\overset{(3.7)(3.14)}{\approx} && 1+\sum_{k=0}^{\infty}(-z)^k \frac{(1-q)^{\alpha k-k}}{\Gamma_q(k+1)}\\
&\overset{(2.9)}{=} && 1+\sum_{k=0}^{\infty}\left(-\frac{z}{(1-q)^{1-\alpha}}\right)^k \frac{1}{[k]_q!}\\
&\overset{(2.3)}{=} && 1+e_q\left(-\frac{z}{(1-q)^{1-\alpha}}\right)\\
&\geq && 1.
\end{aligned}$$

Therefore

$$0<\frac{1}{1+\Gamma_q(1-\alpha)z}\leq e_{\alpha,1}(-z;q), \tag{3.15}$$

and the lower estimates in (3.1) and (3.2) are proved.

The upper estimate. By using (2.1) for $0<\alpha<1$ and $k\in\mathbb{N}$, we have the following inequalities

$$\begin{aligned}
&q^{\alpha+1}/q=q^{\alpha}\leq 1\Rightarrow 1-q\leq 1-q^{\alpha+1} \quad \Rightarrow \quad \prod_{n=0}^{\infty}\left(1-q^{n+1}\right)\leq\prod_{n=0}^{\infty}\left(1-q^{\alpha+1+n}\right)\\
&\overset{(2.1)}{\Rightarrow} \quad 1\leq\frac{(q^{\alpha+1};q)_{\infty}}{(q;q)_{\infty}};
\end{aligned} \tag{3.16}$$

$$\begin{aligned}
&q^{\alpha(k-1)+1}/q=q^{\alpha(k-1)}\leq 1\Rightarrow q^{\alpha(k-1)+1}\leq q \quad \Rightarrow \quad 1-q\leq 1-q^{\alpha(k-1)+1}\\
&\overset{(2.1)}{\Rightarrow} \quad 1\leq\frac{\left(q^{\alpha(k-1)+1};q\right)_{\infty}}{(q;q)_{\infty}};
\end{aligned} \tag{3.17}$$

$$\begin{aligned}
&q^{\alpha k+1}/q^{k+1}=q^{\alpha-1}\geq 1\Rightarrow q^{\alpha k+1}\geq q^{k+1} \quad \Rightarrow \quad 1-q^{\alpha k+1}\leq 1-q^{k+1}\\
&\overset{(2.1)}{\Rightarrow} \quad 1\geq\frac{\left(q^{\alpha k+1};q\right)_{\infty}}{\left(q^{k+1};q\right)_{\infty}}.
\end{aligned} \tag{3.18}$$

We assume that $0<\xi<\alpha$. Then using the inequality $\left(q^{k+1};q\right)_{\infty}\leq 1$, for $k\in\mathbb{N}$, and (3.16) - (3.18), we find that

$$\begin{aligned}
\gamma_{6,q} \quad &:= && \frac{1}{1-\xi}\frac{(1-q)^{\alpha k}}{(\xi q;q)_{\infty}}\frac{(q^{\alpha+1};q)_{\infty}}{(q;q)_{\infty}}\frac{\left(q^{\alpha(k-1)+1};q\right)_{\infty}}{(q;q)_{\infty}}\\
&\overset{(3.16)(3.17)}{\geq} && \frac{1}{1-\xi}\frac{(1-q)^{\alpha k}}{(\xi q;q)_{\infty}}\\
&\geq && \frac{(1-q)^{\alpha k}}{1-\xi}\frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}}\\
&\geq && \frac{(1-q)^{\alpha k}}{1-\alpha}\frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}},
\end{aligned} \tag{3.19}$$

where $\lim_{\xi\to 0}\frac{q^{\xi}}{(\xi q;q)_{\infty}}=1$ and

$$\begin{aligned}\gamma_{7,q} &:= q^{\xi}(1-q)^{\alpha k}\frac{\left(q^{\alpha k+1};q\right)_{\infty}}{(q;q)_{\infty}}\\ &= q^{\xi}(1-q)^{\alpha k}\frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}}\frac{\left(q^{\alpha k+1};q\right)_{\infty}}{(q^{k+1};q)_{\infty}}\\ &\overset{(3.18)}{\leq} q^{\xi}(1-q)^{\alpha k}\frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}}. \qquad (3.20)\end{aligned}$$

From (2.1), (2.9) and (3.19)-(3.20) we have that

$$\begin{aligned}M_q(k) &:= \frac{\Gamma_q^{-1}(\alpha+1)}{\Gamma_q(\alpha(k-1)+1)}-\frac{1}{2\Gamma_q(\alpha k+1)}\\ &\overset{(2.1)}{=} (1-q)^{\alpha k}\frac{(q^{\alpha+1};q)_{\infty}}{(q;q)_{\infty}}\frac{\left(q^{\alpha(k-1)+1};q\right)_{\infty}}{(q;q)_{\infty}}-(1-q)^{\alpha k}\frac{\left(q^{\alpha k+1};q\right)_{\infty}}{(q;q)_{\infty}}\\ &\overset{(3.19)(3.20)}{=} \lim_{\xi\to 0}\left[\gamma_{6,q}-\gamma_{7,q}\right]\\ &\geq (1-q)^{\alpha k}\frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}}\lim_{\xi\to 0}\left[\frac{1}{1-\alpha}-q^{\xi}\right]\\ &= (1-q)^{\alpha k-k}\left[(1-q)^{(k+1)-1}\frac{\left(q^{k+1};q\right)_{\infty}}{(q;q)_{\infty}}\right]\left[\frac{1}{1-\alpha}-1\right]\\ &\overset{(2.9)}{=} \frac{\alpha}{1-\alpha}\frac{(1-q)^{\alpha k-k}}{\Gamma_q(k+1)}. \qquad (3.21)\end{aligned}$$

Thus,

$$\frac{(1-q)^{\alpha k-k}}{\Gamma_q(k+1)} \lesssim M_q(k). \qquad (3.22)$$

Again, for $k\in\mathbb{N}$ using (2.1), we find the following inequalities

$$\begin{aligned}q^{\alpha k+1}/q^{\alpha(k-1)+1}=q^{-\alpha}\geq 1 \quad &\Rightarrow \quad 1-q^{\alpha(k-1)+1}\leq 1-q^{\alpha k+1}\\ &\overset{(2.1)}{\Rightarrow} \left(q^{\alpha(k-1)+1};q\right)_{\infty}\leq\left(q^{\alpha k+1};q\right)_{\infty}; \qquad (3.23)\end{aligned}$$

$$\begin{aligned}q^{\alpha(k-1)+1}/q^{k+1}=q^{-k-\alpha+\alpha k}\geq 1 \quad &\Rightarrow \quad 1-q^{\alpha(k-1)+1}\leq 1-q^{k+1}\\ &\overset{(2.1)}{\Rightarrow} \left(q^{\alpha(k-1)+1};q\right)_{\infty}\leq\left(q^{k+1};q\right)_{\infty}; \qquad (3.24)\end{aligned}$$

$$\begin{aligned}q^{\alpha}/q=q^{\alpha-1}\geq 1 \quad &\Rightarrow \quad 1-q^{\alpha}\leq 1-q\\ &\overset{(2.1)}{\Rightarrow} \frac{(q^{\alpha};q)_{\infty}}{(q;q)_{\infty}}\leq 1. \qquad (3.25)\end{aligned}$$

First, we consider the case $q^\alpha \leq 1-q^\alpha < 1$. Then $\frac{q^\alpha}{1-q^\alpha} \leq 1$. Form (2.1), (3.21) and (3.23)-(3.25), we have that that

$$\begin{aligned}
M_q(k) &\overset{(3.21)}{=} (1-q)^{\alpha k}\frac{(q^{\alpha+1};q)_\infty}{(q;q)_\infty}\frac{\left(q^{\alpha(k-1)+1};q\right)_\infty}{(q;q)_\infty} - (1-q)^{\alpha k}\frac{\left(q^{\alpha k+1};q\right)_\infty}{(q;q)_\infty} \\
&\overset{(3.23)}{\leq} (1-q)^{\alpha k}\left[\frac{(q^{\alpha+1};q)_\infty}{(q;q)_\infty}\frac{\left(q^{\alpha(k-1)+1};q\right)_\infty}{(q;q)_\infty} - \frac{\left(q^{\alpha k-1+1};q\right)_\infty}{(q;q)_\infty}\right] \\
&\leq (1-q)^{\alpha k}\frac{\left(q^{\alpha(k-1)+1};q\right)_\infty}{(q;q)_\infty}\left[\frac{1-q^\alpha}{1-q^\alpha}\frac{(q^{\alpha+1};q)_\infty}{(q;q)_\infty} - 1\right] \\
&\overset{(2.1)(3.24)}{=} (1-q)^{\alpha k}\frac{\left(q^{k+1};q\right)_\infty}{(q;q)_\infty}\left[\frac{1}{1-q^\alpha}\frac{(q^{\alpha};q)_\infty}{(q;q)_\infty} - 1\right] \\
&\overset{(3.25)}{=} (1-q)^{\alpha k}\frac{\left(q^{k+1};q\right)_\infty}{(q;q)_\infty}\left[\frac{1}{1-q^\alpha} - 1\right] \\
&\leq (1-q)^{\alpha k-k}\left[(1-q)^{(k+1)-1}\frac{\left(q^{k+1};q\right)_\infty}{(q;q)_\infty}\right]\left[\frac{1}{1-q^\alpha} - \frac{q^\alpha}{1-q^\alpha}\right] \\
&\overset{(2.9)}{=} \frac{(1-q)^{\alpha k-k}}{\Gamma_q(k+1)}. \qquad (3.26)
\end{aligned}$$

Next, we the case $1-q^\alpha < q^\alpha$. In fact $1/2 < q^\alpha$. Using (3.26) we write that

$$\begin{aligned}
M_q(k) &\leq (1-q)^{\alpha k}\frac{\left(q^{k+1};q\right)_\infty}{(q;q)_\infty}\left[\frac{1}{1-q^\alpha} - 1\right] \\
&\leq (1-q)^{\alpha k}\frac{\left(q^{k+1};q\right)_\infty}{(q;q)_\infty}\left[\frac{1}{1-1/2} - 1\right] \\
&= \frac{(1-q)^{\alpha k-k}}{\Gamma_q(k+1)}.
\end{aligned}$$

Therefore,

$$M_q(k) \lesssim \frac{(1-q)^{\alpha k-k}}{\Gamma_q(k+1)}. \qquad (3.27)$$

According (2.3), (2.9), (3.22) and (3.27) we find that

$$\begin{aligned}
\frac{e_{\alpha,1}(-z;q)}{\left[1+\Gamma_q^{-1}(\alpha+1)z\right]^{-1}} &= e_{\alpha,1}(-z;q) + \Gamma_q^{-1}(\alpha+1)\,z e_{\alpha,1}(-z;q) \\
&\overset{(2.9)}{=} \sum_{k=0}^{\infty}\frac{(-z)^k}{\Gamma_q(\alpha k+1)} + \sum_{k=0}^{\infty}\frac{(-z)^{k+1}\Gamma_q^{-1}(\alpha+1)}{\Gamma_q(\alpha k+1)} \\
&= 1 - \sum_{k=1}^{\infty}\frac{(-z)^k}{\Gamma_q(\alpha k+1)} - \sum_{k=1}^{\infty}\frac{(-z)^k\Gamma_q^{-1}(\alpha+1)}{\Gamma_q(\alpha(k-1)+1)}
\end{aligned}$$

$$
\begin{aligned}
&= 1-\sum_{k=1}^{\infty}(-z)^k\left[\frac{\Gamma_q^{-1}(\alpha+1)}{\Gamma_q(\alpha(k-1)+1)}-\frac{1}{\Gamma_q(\alpha k+1)}\right]\\
&\overset{(3.21)}{=} 1-\sum_{k=1}^{\infty}(-z)^k M_q(k)\\
&\overset{(3.22)(3.27)}{\approx} 1-\sum_{k=0}^{\infty}\frac{\left(-\frac{z}{(1-q)^{1-\alpha}}\right)^k}{[k]_q!}\\
&\overset{(2.3)}{=} 1-e_q\left(-\frac{z}{(1-q)^{1-\alpha}}\right)\\
&\leq 1.
\end{aligned}
$$

Hence,

$$
\begin{aligned}
e_{\alpha,1}(-z;q) &\leq \frac{1}{1+\Gamma_q^{-1}(\alpha+1)z}\\
&\leq \frac{1}{1+\Gamma_q^{-1}(\alpha+1)z}\\
&\leq 1,
\end{aligned}
$$

which implies that also the upper estimates in (3.1) and (3.2) are proved. This completes the proof. □

Lemma 3.2. *Let $0<\alpha<2$, $0\leq\beta$ and $z>0$. Then we get*

$$
|e_{\alpha,\beta}(-z;q)|\leq\frac{C_q}{1+z}, \tag{3.28}
$$

where $C_q:=\frac{e_{q^{1/2}}(\alpha')}{\Gamma_q(\beta)}$ and $\alpha':=\frac{2^\alpha}{(1-q^\beta)(1-q^{1/2})}$.

Proof. Let $0<\alpha<2$, $0\leq\beta$ and $k\in\mathbb{N}$. From (2.1), we get the following inequalities

$$
\begin{aligned}
q^{2k+\beta}\leq q^{\alpha k+\beta}\leq q^{\alpha(k-1)+\beta} &\Rightarrow 1-q^{\alpha(k-1)+\beta}\leq 1-q^{\alpha k+\beta}\leq 1-q^{2k+\beta}\\
&\overset{(2.1)}{\Rightarrow} \left(q^{\alpha(k-1)+\beta};q\right)_\infty\leq\left(q^{\alpha k+\beta};q\right)_\infty\leq\left(q^{2k+\beta};q\right)_\infty;\\
&\Rightarrow (1-q)^{\alpha k}\leq(1-q)^{\alpha(k-1)}, \qquad (3.29)
\end{aligned}
$$

and

$$
\begin{aligned}
q^{\beta+1}\leq q\leq q^{1/2} &\overset{(2.1)}{\Rightarrow} \left(q^{1/2};q^{1/2}\right)_k\leq\left(q^{\beta+1};q^2\right)_k;\\
(q^\beta;q^2)_k &\overset{(2.1)}{=} (1-q^\beta)(1-q^{\beta+2})\dots(1-q^{\beta+k-2})\\
&\Rightarrow (1-q^\beta)^k\leq(q^\beta;q^2)_k. \qquad (3.30)
\end{aligned}
$$

Thus, using (2.2), (3.29) and (3.30) we find that

$$
\begin{aligned}
\gamma_{8,q} &:= \frac{1}{\Gamma_q(\alpha k+\beta)}+\frac{1}{\Gamma_q(\alpha(k-1)+\beta)}\\
&\overset{(2.9)}{=} (1-q)^{\alpha k+\beta-1}\frac{\left(q^{\alpha k+\beta};q\right)_\infty}{(q;q)_\infty}\\
&+ (1-q)^{\alpha(k-1)+\beta-1}\frac{\left(q^{\alpha(k-1)+\beta};q\right)_\infty}{(q;q)_\infty}\\
&\overset{(3.29)}{\leq} 2(1-q)^{\alpha(k-1)}\left[(1-q)^{\beta-1}\frac{\left(q^{\beta};q\right)_\infty}{(q;q)_\infty}\right]\frac{\left(q^{2k+\beta};q\right)_\infty}{\left(q^{\beta};q\right)_\infty}\\
&\overset{(2.9)(2.2)}{=} \frac{2}{\Gamma(\beta)}(1-q)^{\alpha(k-1)}\frac{1}{\left(q^{\beta};q\right)_{2k}}\\
&\overset{(2.2)}{=} \frac{2}{\Gamma(\beta)}(1-q^{1/2})^{\alpha(k-1)}\frac{(1+q^{1/2})^{\alpha(k-1)}}{\left(q^{\beta};q^2\right)_k\left(q^{\beta+1};q^2\right)_k}\\
&\overset{(3.30)}{=} \frac{1}{\Gamma(\beta)}(1-q^{1/2})^{\alpha(k-1)}\frac{2^{\alpha k}}{\left(1-q^{\beta}\right)^k\left(q^{1/2};q^{1/2}\right)_k}\\
&= \frac{1}{\Gamma(\beta)}(1-q^{1/2})^{\alpha(k-1)}\frac{\left[\frac{2^{\alpha}}{\left(1-q^{\beta}\right)(1-q^{1/2})}\right]^k}{\left[\frac{\left(q^{1/2};q^{1/2}\right)_k}{(1-q^{1/2})^k}\right]}\\
&= \frac{1}{\Gamma(\beta)}(1-q^{1/2})^{\alpha(k-1)}\frac{(\alpha')^k}{[k]_{q^{1/2}}!}, \qquad (3.31)
\end{aligned}
$$

where $\alpha'=\frac{2^{\alpha}}{\left(1-q^{\beta}\right)(1-q^{1/2})}$.

Thus, form (2.1), (3.29) and (3.31) we find that

$$
\begin{aligned}
\frac{e_{\alpha,\beta}\left(-z;q\right)}{[1+z]^{-1}} &\overset{(1.4)}{=} \sum_{m=0}^{\infty}\frac{(-z)^k}{\Gamma_q(\alpha k+\beta)}+z\sum_{m=0}^{\infty}\frac{(-z)^k}{\Gamma_q(\alpha k+\beta)}\\
&\leq \frac{1}{\Gamma_q(\beta)}+\sum_{m=1}^{\infty}\frac{z^k}{\Gamma_q(\alpha k+\beta)}+\sum_{m=1}^{\infty}\frac{z^k}{\Gamma_q(\alpha(k-1)+\beta)}\\
&= \frac{1}{\Gamma_q(\beta)}+\sum_{m=1}^{\infty}z^k\left[\frac{1}{\Gamma_q(\alpha k+\beta)}+\frac{1}{\Gamma_q(\alpha(k-1)+\beta)}\right]\\
&\overset{(3.31)}{=} \frac{1}{\Gamma_q(\beta)}+\sum_{m=1}^{\infty}z^k\gamma_{8,q}\\
&\overset{(3.31)}{\leq} \frac{1}{\Gamma_q(\beta)}+\frac{1}{\Gamma(\beta)}\sum_{m=1}^{\infty}\left[(1-q^{1/2})^{\frac{\alpha(k-1)}{k}}z\right]^k\frac{(\alpha')^k}{[k]_{q^{1/2}}!}
\end{aligned}
$$

$$\begin{aligned} &\leq \frac{1}{\Gamma_q(\beta)} + \frac{1}{\Gamma(\beta)} \sum_{m=1}^{\infty} \frac{(\alpha')^k}{[k]_{q^{1/2}}!} \\ &\leq \frac{1}{\Gamma_q(\beta)} \left[1 + \sum_{m=1}^{\infty} \frac{(\alpha')^k}{[k]_{q^{1/2}}!} \right] \\ &\leq \frac{e_{q^{1/2}}(\alpha')}{\Gamma_q(\beta)}. \end{aligned}$$

Therefore,

$$|e_{\alpha,\beta}(-z;q)| \leq \frac{e_{q^{1/2}}(\alpha')}{\Gamma_q(\beta)} \frac{1}{1+z},$$

which means that the estimate (3.28) holds. The proof is complete. □

4. Direct problem

In this section we consider the Cauchy problem for the time-fractional subdiffusion in the quantum calculus. We study existence and uniqueness results for this problem, based on the $\mathcal{L}$–Fourier method. An introduction and some basic definitions of the $\mathcal{L}$-Fourier analysis are given in [20], [31], [33] and references therein. We briefly describe the definitions used in this paper, the global Fourier analysis that has been developed in [32]. Let H be a separable Hilbert space and $\mathcal{L}$ be a positive operator with the corresponding discrete spectrum $\{\lambda_k\}_{k\in I}$ on H, so that $\lambda_k \in \mathbb{R}$: $\lambda_k \geq 0$ for all $k \in I$, where I is a countable set ($I = \mathbb{N}^k$ or $I = \mathbb{Z}^k$ for some k). The Plancherel identity on H takes the form

$$\|u\|_H = \left(\sum_{k\in I} |\langle u, \psi_k \rangle_H|^2 \right)^{\frac{1}{2}}, \tag{4.1}$$

for $u \in H$, where ψ_k is an orthonormal basis of H corresponding to λ_k.

Consequently, we can also define the Sobolev spaces $\mathcal{H}^d_{\mathcal{L}}$ associated to $\mathcal{L}$ as

$$\mathcal{H}^d_{\mathcal{L}} := \left\{ u \in H : (I+\mathcal{L})^{d/2} u \in H \right\}$$

for any $d \in \mathbb{R}$. Using Plancherel's identity, we can write the norm in the following form:

$$\|u\|_{\mathcal{H}^d_{\mathcal{L}}} := \|(I+\mathcal{L})^{d/2} u\|_H = \left(\sum_{k\in I} (1+\lambda_k)^d |\langle u, \psi_k \rangle_H|^2 \right)^{\frac{1}{2}}. \tag{4.2}$$

For $u :\in [0,T] \to \mathcal{H}^d_{\mathcal{L}}$ we introduce the spaces $W^{\alpha}_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right)$, $C^m_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right)$ and $L^{\infty}_q[0,T]$ as

$$\|u\|_{W^{\alpha}_q([0,T];\mathcal{H}^d_{\mathcal{L}})} := \max_{0\leq t\leq T} \|u(t)\|_{\mathcal{H}^d_{\mathcal{L}}} + \max_{0\leq t\leq T} \|{}^cD^{\alpha}_q u(t)\|_{\mathcal{H}^d_{\mathcal{L}}}$$

for $0 < \alpha < 2$, and

$$\|u\|_{C^m_q([0,T];\mathcal{H}^d_{\mathcal{L}})} := \sum_{i=0}^{m} \sup_{0<t\leq T} \|D^i_q u(t)\|_{\mathcal{H}^d_{\mathcal{L}}},$$

and

$$\|u\|_{L_q^\infty([0,T];\mathcal{H}_{\mathcal{L}}^d)} = \sup_{0\le t\le T}\|u(t)\|_{\mathcal{H}_{\mathcal{L}}^d}.$$

4.1. **The $0<\alpha\le 1$ case.** The first purpose of this section is to study the direct time-fractional subdiffusion problem for the equation

$$ {}^cD_q^\alpha u(t) + \mathcal{L}u(t) + mu(t) = f(t)\in H,\ \ t>0, \tag{4.3}$$

with the initial data

$$u(0)=\varphi\in H, \tag{4.4}$$

where ${}^cD^\alpha_{0+,t}u(t)$ is the Caputo fractional q-derivatives of order α of $u(t)$ with respect to t (see (2.10) and $m>0$.

A generalised solution of Problem (4.3)-(4.4) is a function

$$u\in L_q^\infty\left([0,T];\mathcal{H}_{\mathcal{L}}^{d+2}\right)\cap W_q^\alpha\left([0,T];\mathcal{H}_{\mathcal{L}}^{d}\right).$$

For this considered Problem, the following theorem holds true.

Theorem 4.1. *Let $0<\alpha\le 1$, $0<T<\infty$, $\varphi\in\mathcal{H}_{\mathcal{L}}^{d+2}$ and $f\in C_q^1\left([0,T];\mathcal{H}_{\mathcal{L}}^{d}\right)$. Then there exists a unique solution $u(t)$ of Problem (4.3)-(4.4) such that*

$$u\in L_q^\infty\left([0,T];\mathcal{H}_{\mathcal{L}}^{d+2}\right)\cap W_q^\alpha\left([0,T];\mathcal{H}_{\mathcal{L}}^{d}\right).$$

Moreover, this solution can be written in the form

$$\begin{aligned} u(t) &= \sum_{k\in I}\left[\varphi_k e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^\alpha;q\right)\right]\psi_k \\ &+ \sum_{k\in I}\left[\int_0^t t^{\alpha-1}\left(qs/t;q\right)_{\alpha-1}\varepsilon^{-q^\alpha s}e_{\alpha,\alpha}\left(-\left(\lambda_k+m\right)t^\alpha;q\right)f_k(s)d_qs\right]\psi_k, \end{aligned}$$

which satisfies the estimate

$$\|{}^cD_q^\alpha u(t)\|^2_{\mathcal{H}_{\mathcal{L}}^d}+\|u(t)\|^2_{\mathcal{H}_{\mathcal{L}}^{d+2}}\le C_T\left[\|\varphi\|^2_{\mathcal{H}_{\mathcal{L}}^{d+2}}+\|f\|^2_{C_q^1([0,T];\mathcal{H}_{\mathcal{L}}^d)}\right],\quad 0<t\le T,$$

where the q-translation operator $\varepsilon^{-q^\alpha s}$ is defined in (2.6) and $C_T:=\max\{2,T\}$.

Proof. Existence. Since the system of eigenfunctions ψ_k is a basis in H, we seek a generalised solution by $u(t)$ in the form

$$u(t)=\sum_{k\in I}u_k(t)\psi_k, \tag{4.5}$$

and we represent the function on the right-hand side of the equation (4.2) in the form:

$$f(t)=\sum_{k\in I}f_k(t)\psi_k, \tag{4.6}$$

where the Fourier coefficients are defined by the formula

$$u_k(t):=\langle u(t),\psi_k\rangle_H,\quad f_k(t):=\langle f(t),\psi_k\rangle_H.$$

Moreover, we find the expression for the derivatives of (4.5) -(4.6):

$$ {}^{c}D_{q}^{\alpha}u(t) = \sum_{k\in I} {}^{c}D_{q}^{\alpha}u_{k}(t)\psi_{k}, \tag{4.7} $$

$$ \mathcal{L}u(t) = \sum_{k\in I} \lambda_{k}u_{k}(t)\psi_{k}. \tag{4.8} $$

Then substituting (4.6) -(4.8) into (4.3) and (4.4), we get

$$ {}^{c}D_{q,t}^{\alpha}u_{k}(t) + (\lambda_{k}+m)\,u_{k}(t) = f_{k}(t), \tag{4.9} $$

$$ u_{k}(0) = \varphi_{k}, \tag{4.10} $$

for $k \in I$ and $t > 0$.

A generalised solution of the problem (4.8) - (4.10) ([35, Theorem 3.1]) is given by

$$ \begin{aligned} u_{k}(t) &= \varphi_{k}e_{\alpha,1}\left(-(\lambda_{k}+m)\,t^{\alpha};q\right) \\ &+ \int_{0}^{t} t^{\alpha-1}\left(qs/t;q\right)_{\alpha-1}\varepsilon^{-q^{\alpha}s}e_{\alpha,\alpha}\left(-(\lambda_{k}+m)\,t^{\alpha};q\right)f_{k}(s)d_{q}s. \end{aligned} \tag{4.11} $$

This, combined with (4.5) and (4.11) gives

$$ \begin{aligned} u(t) &= \sum_{k\in I}\left[\varphi_{k}e_{\alpha,1}\left(-(\lambda_{k}+m)\,t^{\alpha};q\right)\right]\psi_{k} \\ &+ \sum_{k\in I}\left[\int_{0}^{t} t^{\alpha-1}\left(qs/t;q\right)_{\alpha-1}\varepsilon^{-q^{\alpha}s}e_{\alpha,\alpha}\left(-(\lambda_{k}+m)\,t^{\alpha};q\right)f_{k}(s)d_{q}s\right]\psi_{k}. \end{aligned} $$

Convergence. We assume that $i = 0, 1, 2\ldots$, and $0 < \alpha \leq 1$. Then by following (see [9, Proposition 14.1]) we have

$$ t^{\alpha i-1}(q^{\alpha}s/t;q)_{\alpha i-1} = t^{\alpha(i-1)}(q^{\alpha}s;q)_{\alpha(i-1)}\left[t^{\alpha-1}(qs/t;q)_{\alpha-1}\right]. \tag{4.12} $$

For a fixed k, let us denote $\lambda_{0} := \lambda_{k} + m$ for $m > 0$. Then by making use of (2.5), (2.6), (2.9) and (4.12), we have

$$\begin{aligned}
D_{q,s}\left[\varepsilon^{-q^{\alpha}s}e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)\right] &= D_{q,s}\left[e_{\alpha,1}\left(-\lambda_0 t^{\alpha}(q^{\alpha}s/t;q)_{\alpha};q\right)\right]\\
&\overset{(2.6)}{=} \sum_{i=1}^{\infty}\frac{(-\lambda_0)^i}{\Gamma_q(\alpha i+1)}D_{q,s}\left[t^{\alpha i}(q^{\alpha}s/t;q)_{\alpha i}\right]\\
&\overset{(2.5)}{=} -\sum_{i=1}^{\infty}(-\lambda_0)^i\frac{[\alpha i]_q}{\Gamma_q(\alpha i+1)}t^{\alpha i-1}(q^{\alpha}s/t;q)_{\alpha i-1}\\
&\overset{(2.9)}{=} -\sum_{i=1}^{\infty}\frac{(-\lambda_0)^i}{\Gamma_q(\alpha i)}t^{\alpha i-1}(q^{\alpha}s/t;q)_{\alpha i-1}\\
&\overset{(4.12)}{=} \lambda_0\sum_{i=1}^{\infty}\frac{(-\lambda_0)^{i-1}}{\Gamma(\alpha i)}\left[\frac{t^{\alpha(i-1)}(q^{\alpha}s;q)_{\alpha(i-1)}}{\left[t^{\alpha-1}(qs/t;q)_{\alpha-1}\right]^{-1}}\right]\\
&= \lambda_0\frac{\sum\limits_{i=0}^{\infty}\frac{(-\lambda_k)^i t^{\alpha i}(q^{\alpha}s/t;q)_{\alpha i}}{\Gamma(\alpha i+\alpha)}}{\left[t^{\alpha-1}(qs/t;q)_{\alpha-1}\right]^{-1}}\\
&= \lambda_0\frac{\varepsilon^{-q^{\alpha}s}e_{\alpha,\alpha}\left(-\lambda_0 t^{\alpha};q\right)}{\left[t^{\alpha-1}(qs/t;q)_{\alpha-1}\right]^{-1}}.
\end{aligned}$$

Thus,

$$\frac{\varepsilon^{-q^{\alpha}s}e_{\alpha,\alpha}\left(-\lambda_0 t^{\alpha};q\right)}{\left[t^{\alpha-1}(qs/t;q)_{\alpha-1}\right]^{-1}} = \frac{D_{q,s}\left[\varepsilon^{-q^{\alpha}s}e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)\right]}{\lambda_0}. \tag{4.13}$$

Form (2.8) and (4.13) we get that

$$\begin{aligned}
\int_0^t\frac{\varepsilon^{-q^{\alpha}s}e_{\alpha,\alpha}\left(-\lambda_0 t^{\alpha};q\right)f_k(s)}{\left[t^{\alpha-1}\left(qs/t;q\right)^{\alpha-1}\right]^{-1}}d_qs &\overset{(4.13)}{=} \int_0^t\frac{D_{q,s}\left[\varepsilon^{-q^{\alpha}s}e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)\right]f_k(s)d_qs}{\lambda_0}\\
&\overset{(2.8)}{=} \frac{f_k(t)\varepsilon^{-q^{\alpha}t}e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)}{\lambda_0} - \frac{f_k(0)e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)}{\lambda_0}\\
&- \int_0^t\frac{\varepsilon^{-q^{\alpha}s}e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)D_qf_k(s)d_qs}{\lambda_0}.
\end{aligned} \tag{4.14}$$

Since $\varphi\in\mathcal{H}_{\mathcal{L}}^{d+2}$, $f\in C_q^1\left([0,T];\mathcal{H}_{\mathcal{L}}^d\right)$, then from (3.28), (4.11) and (4.14) it follows that

$$\begin{aligned}
\left|\langle u(t),\psi_k\rangle_H\right| &\overset{(4.11)}{\leq} \left|\langle\varphi(t),\psi_k\rangle_H\right|\left|e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)\right|\\
&+ \left|\int_0^t t^{\alpha-1}\left(qs/t;q\right)_{\alpha-1}\varepsilon^{-q^{\alpha}s}e_{\alpha,\alpha}\left(-\lambda_0 t^{\alpha};q\right)\langle f(t),\psi_k\rangle_H d_qs\right|\\
&\overset{(4.14)}{\leq} \left|\langle\varphi(t),\psi_k\rangle_H\right|\left|e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)\right| + \frac{\left|\langle f(t),\psi_k\rangle_H\right|\left|\varepsilon^{-q^{\alpha}t}e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)\right|}{\lambda_0}\\
&+ \frac{\left|e_{\alpha,1}\left(-\lambda_0 t^{\alpha};q\right)\right|\left|\langle f(0),\psi_k\rangle_H\right|}{\lambda_0}
\end{aligned}$$

$$
\begin{aligned}
&+ \quad \frac{\int_0^t \left|\varepsilon^{-q^\alpha s} e_{\alpha,1}\left(-\lambda_0 t^\alpha; q\right) \langle D_q f(t), \psi_k\rangle_H\right| d_q s}{\lambda_0} \\
&\overset{(3.28)}{\lesssim} \quad |\langle \varphi(t), \psi_k\rangle_H| + \frac{|\langle f(t), \psi_k\rangle_H|}{\lambda_0} \\
&+ \quad \frac{|\langle f(0), \psi_k\rangle_H|}{\lambda_0} + \frac{1}{\lambda_0}\int_0^t |D_q\langle f(t), \psi_k\rangle_H|\, d_q s \\
&\leq \quad |\langle \varphi, \psi_k\rangle_H| + \frac{2}{\lambda_0}\sup_{0<t\leq T}|\langle f(t), \psi_k\rangle_H| + \frac{T}{\lambda_0}\sup_{0<t\leq T}|\langle D_q f(t), \psi_k\rangle_H|,
\end{aligned}
$$

which implies

$$
\begin{aligned}
(1+\lambda_k)^{d/2}\,|\langle u(t), \psi_k\rangle_H| \;\leq\; & C_T\left[\left|\langle (1+\lambda_k)^{d/2}\,\varphi, \psi_k\rangle_H\right|\right. \\
& + \sup_{0<t\leq T}\left|\langle (1+\lambda_k)^{d/2-1}\, f(t), \psi_k\rangle_H\right| \\
& \left. + \sup_{0<t\leq T}\left|\langle (1+\lambda_k)^{d/2-1}\, D_q f(t), \psi_k\rangle_H\right|\right], \qquad (4.15)
\end{aligned}
$$

where $C_T := \max\{2, T\}$.

By the definition of the Fourier coefficient and (4.9) and (4.15), we find that

$$
\begin{aligned}
(1+\lambda_k)^{d/2}\left|\langle {}^cD_q^\alpha u(t), \psi_k\rangle_H\right| &= (1+\lambda_k)^{d/2}\,|{}^cD_q^\alpha u_k(t)| \\
&\overset{(4.9)}{\lesssim} (1+\lambda_k)^{d/2+1}\,|u_k(t)| + (1+\lambda_k)^{d/2}\,|f_k(t)| \\
&\overset{(4.15)}{\leq} C_T\left[\left|\langle (1+\lambda_k)^{d/2+1}\,\varphi, \psi_k\rangle_H\right|\right. \\
&\quad + \sup_{0<t\leq T}\left|\langle (1+\lambda_k)^{d/2}\, f(t), \psi_k\rangle_H\right| \\
&\quad \left. + \sup_{0<t\leq T}\left|\langle (1+\lambda_k)^{d/2}\, D_q f(t), \psi_k\rangle_H\right|\right] \\
&= C_T\left[\left|\langle (I+\mathcal{L})^{d/2+1}\,\varphi, \psi_k\rangle_H\right|\right. \\
&\quad + \sup_{0<t\leq T}\left|\langle (I+\mathcal{L})^{d/2}\, f(t), \psi_k\rangle_H\right| \\
&\quad \left. + \sup_{0<t\leq T}\left|\langle (I+\mathcal{L})^{d/2}\, D_q f(t), \psi_k\rangle_H\right|\right], \qquad (4.16)
\end{aligned}
$$

and

$$\begin{aligned}(1+\lambda_k)^{d/2}\left|\langle \mathcal{L}u(t),\psi_k\rangle_H\right| &= (1+\lambda_k)^{d/2}\left|\langle \lambda_k u(t),\psi_k\rangle_H\right| \\ &\overset{(4.15)}{\leq} C_T\left[\left|\left\langle (1+\lambda_k)^{d/2+1}\varphi,\psi_k\right\rangle_H\right|\right. \\ &+ \sup_{0<t\leq T}\left|\left\langle (1+\lambda_k)^{d/2} f(t),\psi_k\right\rangle_H\right| \\ &+ \left.\sup_{0<t\leq T}\left|\left\langle (1+\lambda_k)^{d/2} D_q f(t),\psi_k\right\rangle_H\right|\right] \\ &= C_T\left[\left|\left\langle (I+\mathcal{L})^{d/2+1}\varphi,\psi_k\right\rangle_H\right|\right. \\ &+ \sup_{0<t\leq T}\left|\left\langle (I+\mathcal{L})^{d/2} f(t),\psi_k\right\rangle_H\right| \\ &+ \left.\sup_{0<t\leq T}\left|\left\langle (I+\mathcal{L})^{d/2} D_q f(t),\psi_k\right\rangle_H\right|\right]. \end{aligned} \tag{4.17}$$

Using (4.2), (4.16), (4.17), and the Parseval identity, we get that

$$\begin{aligned}\|{}^cD_q^\alpha u(t)\|^2_{\mathcal{H}^d_{\mathcal{L}}} &\overset{(4.2)}{=} \sum_{k\in I}(1+\lambda_k)^d\left|\langle {}^cD_q^\alpha u(t),\psi_k\rangle_H\right|^2 \\ &\overset{(4.16)}{\leq} C_T\left\{\sum_{k\in I}\left|\left\langle (I+\mathcal{L})^{d/2+1}\varphi,\psi_k\right\rangle_H\right|^2\right. \\ &+ \sum_{k\in I}\left[\sup_{0<t\leq T}\left|\left\langle (I+\mathcal{L})^{d/2} f(t),\psi_k\right\rangle_H\right|^2\right. \\ &+ \left.\left.\sup_{0<t\leq T}\left|\left\langle (I+\mathcal{L})^{d/2} D_q f(t),\psi_k\right\rangle_H\right|^2\right]\right\} \\ &\overset{(4.2)}{=} C_T\left[\left\|\left(I+\mathcal{L}\right)^{d/2+1}\varphi\right\|^2_H + \sup_{0<t\leq T}\left\|\left(I+\mathcal{L}\right)^{d/2} f(t)\right\|^2_H\right. \\ &+ \left.\sup_{0<t\leq T}\left\|\left(1+\mathcal{L}\right)^{d/2} D_q f(t)\right\|^2_H\right] \\ &= C_T\left[\|\varphi\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} + \|f\|^2_{C^1_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right)}\right] < \infty, \end{aligned}$$

and

$$\begin{aligned}\left\|\left(I+\mathcal{L}\right)u(t)\right\|^2_{\mathcal{H}^d_{\mathcal{L}}} &\overset{(4.2)}{=} \sum_{k\in I}\left|(1+\lambda_k)^{d/2}\left\langle \left(I+\mathcal{L}\right)u(t),\psi_k\right\rangle_H\right|^2 \\ &\overset{(4.17)}{\leq} C_T\left[\|\varphi\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} + \|f\|^2_{C^1_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right)}\right] < \infty. \end{aligned}$$

Hence, the above estimates imply that

$$\|{}^cD_q^\alpha u(t)\|^2_{\mathcal{H}^d_{\mathcal{L}}} + \|u(t)\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} \leq C_T\left[\|\varphi\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} + \|f\|^2_{C^1_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right)}\right],$$

which means that $u\in L^\infty_q\left([0,T];\mathcal{H}^{d+2}_{\mathcal{L}}\right)\cap W^\alpha_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right)$.

The uniqueness. We assume that there are two solutions $v_1(t)$ and $v_2(t)$ of the problem (4.2)-(4.3). Then we must have that, for $w = u - v$,

$$\begin{aligned} {}^cD_q^\alpha w + \mathcal{L}w &= {}^cD_q^\alpha (u-v) + \mathcal{L}(u-v) + mw(u-v) \\ &= {}^cD_t^\alpha u + \mathcal{L}u - \left[{}^cD_q^\alpha v + \mathcal{L}v\right] \\ &= f - f = 0, \end{aligned}$$

with the initial conditions

$$w(0) = u(0) - v(0) = \varphi - \varphi = 0.$$

Therefore, $w = 0$, and the solution must be unique. The proof is complete. □

4.2. **The $1 < \alpha < 2$ case.** This subsection is concerned with a Cauchy problem for the time-fractional subdiffusion equation

$$ {}^cD_q^\alpha u(t) + \mathcal{L}u(t) + mu(t) = f(t) \in H, \tag{4.18}$$

with the initial data

$$u(0) = \varphi \in H, \quad D_q u(0) = \rho \in H. \tag{4.19}$$

Theorem 4.2. *Let $1 < \alpha < 2$, $m > 0$ and $0 < T < \infty$. We assume that $\varphi, \rho \in \mathcal{H}_{\mathcal{L}}^{d+2}$ and $f \in C_q^1\left([0,T]; \mathcal{H}_{\mathcal{L}}^d\right)$. Then there exists a unique solution of Problem (4.18)-(4.19):*

$$u \in L_q^\infty\left([0,T]; \mathcal{H}_{\mathcal{L}}^{d+2}\right) \cap W_q^\alpha\left([0,T]; \mathcal{H}_{\mathcal{L}}^d\right).$$

Moreover, this solution can be written in the form

$$\begin{aligned} u(t) &= \sum_{k\in I} \Big[\varphi_k e_{\alpha,1}\left(-(\lambda_k+m)t^\alpha; q\right) + t\rho_k e_{\alpha,2}\left(-(\lambda_k+m)t^\alpha; q\right) \\ &+ \int_0^t t^{\alpha-1}\left(qs/t; q\right)_{\alpha-1} \varepsilon^{-q^\alpha s} e_{\alpha,\alpha}\left(-(\lambda_k+m)t^\alpha; q\right) f_k(s) d_q s\Big] \phi_k, \end{aligned} \tag{4.20}$$

which satisfies the estimate

$$\|{}^cD_q^\alpha u(t)\|_{\mathcal{H}_{\mathcal{L}}^d}^2 + \|u(t)\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^2 \le C_T\left[\|\varphi\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^2 + \|\rho\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^2 + \|f\|_{C_q^1\left([0,T];\mathcal{H}_{\mathcal{L}}^d\right)}^2\right], \tag{4.21}$$

for $0 \le t \le T$.

Proof. Existence. By repeating the arguments in the proof of Theorem 4.1, we have the equation (4.7) with the initial conditions

$$u_k(0) = \varphi_k, \quad D_q u_k(0) = \rho_k, \quad k \in I, \tag{4.22}$$

and a generalised solution in the following form:

$$\begin{aligned} u_k(t) &= \varphi_k e_{\alpha,1}\left(-(\lambda_k+m)t^\alpha; q\right) + t\rho_k e_{\alpha,2}\left(-(\lambda_k+m)t^\alpha; q\right) \\ &+ \int_0^t t^{\alpha-1}\left(qs/t; q\right)_{\alpha-1} \varepsilon^{-q^\alpha s} e_{\alpha,\alpha}\left(-(\lambda_k+m)t^\alpha; q\right) f_k(s) d_q s. \end{aligned} \tag{4.23}$$

Using (4.5) and (4.23) we obtain the solution (4.20).

Convergence. By using (4.14) and (4.23), we get that

$$\begin{aligned} u_k(t) &\overset{(4.14)(4.23)}{=} \varphi_k e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^\alpha;q\right)+t\rho_k e_{\alpha,2}\left(-\left(\lambda_k+m\right)t^\alpha;q\right)\\ &+ \frac{1}{\lambda_k+m}f_k(t)e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^\alpha(q^\alpha;q)_\alpha;q\right)\\ &- \frac{1}{\lambda_k+m}f_k(0)e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^\alpha;q\right)\\ &- \frac{1}{\lambda_k+m}\int_0^t \varepsilon^{-q^{\alpha+1}s}e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^\alpha;q\right)D_{q,s}f_k(s)d_qs. \end{aligned} \tag{4.24}$$

From (3.28) and (4.24) we conclude that

$$\begin{aligned} \left(1+\lambda_k\right)^{d/2}\left|\langle u(t),\psi_k\rangle_H\right| &\overset{(3.28)(4.24)}{\leq} C_T\Big\{\left(1+\lambda_k\right)^{d/2}\left|\langle\varphi,\psi_k\rangle_H\right|\\ &+ \left(1+\lambda_k\right)^{d/2}\left|\langle\rho,\psi_k\rangle_H\right|\\ &+ \left(1+\lambda_k\right)^{d/2-1}\sup_{0<s\leq T}\left|\langle f(t,\cdot),\psi_k\rangle_H\right|\\ &+ \left(1+\lambda_k\right)^{d/2-1}\sup_{0<s\leq T}\left|\langle D_qf(s,\cdot),\psi_k\rangle_H\right|\Big\}, \end{aligned} \tag{4.25}$$

where $C_T:=\max\{2,T\}$.

Since $\varphi,\rho\in\mathcal{H}_{\mathcal{L}}^{d+2}$, $f\in C_q^1\left([0,T];\mathcal{H}_{\mathcal{L}}^d\right)$, using (4.9) and (4.25) we have that

$$\begin{aligned} \left(1+\lambda_k\right)^{d/2}\left|\langle {}^cD_q^\alpha u(t),\psi_k\rangle_H\right| &\overset{(4.9)}{\lesssim} \left(1+\lambda_k\right)^{d/2}\left|u_k(t)\right|+\left|f_k(t)\right|\\ &\overset{(4.25)}{\leq} C_T\Big\{\left|\left\langle\left(I+\mathcal{L}\right)^{d/2+1}\varphi,\psi_k\right\rangle_H\right|\\ &+ \left|\left\langle\left(I+\mathcal{L}\right)^{d/2+1}\rho,\psi_k\right\rangle_H\right|\\ &+ \sup_{0<s\leq T}\left|\left\langle\left(I+\mathcal{L}\right)^{d/2}f(t),\psi_k\right\rangle_H\right|\\ &+ \sup_{0<s\leq T}\left|\left\langle\left(I+\mathcal{L}\right)^{d/2}D_qf(t),\psi_k\right\rangle_H\right|\Big\}, \end{aligned} \tag{4.26}$$

and

$$\begin{aligned} \left(1+\lambda_k\right)^{d/2}\left|\left\langle\left(I+\mathcal{L}\right)u(t),\psi_k\right\rangle_H\right| &\overset{4.25)}{\leq} C_T\Big\{\left|\left\langle\left(I+\mathcal{L}\right)^{d/2+1}\varphi,\psi_k\right\rangle_H\right|\\ &+ \left|\left\langle\left(I+\mathcal{L}\right)^{d/2+1}\rho,\psi_k\right\rangle_H\right|\\ &+ \sup_{0<s\leq T}\left|\left\langle\left(I+\mathcal{L}\right)^{d/2}f(t),\psi_k\right\rangle_H\right|\\ &+ \sup_{0<s\leq T}\left|\left\langle\left(I+\mathcal{L}\right)^{d/2}D_qf(t),\psi_k\right\rangle_H\right|\Big\}. \end{aligned} \tag{4.27}$$

According (4.26) and (4.27) we get (4.21).

Hence, the above estimates imply that

$$\|{}^cD_q^\alpha u(t)\|^2_{\mathcal{H}^d_{\mathcal{L}}} + \|u(t)\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} \leq C_T \left[\|\varphi\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} + \|\rho\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} + \|f\|^2_{C^1_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right)}\right],$$

which means that $u \in L^\infty_q\left([0,T];\mathcal{H}^{d+2}_{\mathcal{L}}\right) \cap W^\alpha_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right)$.

Uniqueness. This part can be proved similarly to the proof of Theorem 4.1, so we omit the details. The proof is complete. □

5. Inverse problem

The subsection deals with an inverse problem concerning the time-fractional diffusion equation. The problem is to find the couple $(u(t), f)$ satisfying the equation:

$$ {}^cD^\alpha_{q,t}u(t) + \mathcal{L}u(t) + mu(t) = f \in H, \quad t > 0, \tag{5.1}$$

with initial data

$$u(0) = \varphi \in H, \tag{5.2}$$

and final condition:

$$u(T) = \rho \in H, \tag{5.3}$$

where $m > 0$.

Theorem 5.1. *Let $0 < \alpha < 1$ and $0 < T < \infty$. Assume that $\varphi, \rho \in \mathcal{H}^{d+2}_{\mathcal{L}}$. Then the generalised solution of (5.1)-(5.3), exists, is unique, and can be written in the form*

$$u(t) = \varphi + \sum_{k\in I} \frac{[\varphi_k - \rho_k]\left[e_{\alpha,1}\left(-\left(\lambda_k + m\right)t^\alpha; q\right) - 1\right]\psi_k}{\left[1 - e_{\alpha,1}\left(-\left(\lambda_k + m\right)T^\alpha; q\right)\right]},$$

and

$$f = \mathcal{L}\varphi - \sum_{k\in I} \frac{\left(\lambda_k + m\right)[\varphi_k - \rho_k]\,\psi_k}{1 - e_{\alpha,1}\left(-\left(\lambda_k + m\right)T^\alpha; q\right)}.$$

Moreover,

$$u \in L^\infty_q\left([0,T];\mathcal{H}^{d+2}_{\mathcal{L}}\right) \cap W^\alpha_q\left([0,T];\mathcal{H}^d_{\mathcal{L}}\right); \quad f \in \mathcal{H}^d_{\mathcal{L}},$$

which satisfy the estimates

$$\begin{aligned} \|{}^cD^\alpha_q u(t)\|^2_{\mathcal{H}^d_{\mathcal{L}}} &+ \|u(t)\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} \lesssim \|\varphi\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} + \|\rho\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}}; \\ \|f\|^2_{\mathcal{H}^d_{\mathcal{L}}} &\lesssim \|\varphi\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}} + \|\rho\|^2_{\mathcal{H}^{d+2}_{\mathcal{L}}}. \end{aligned} \tag{5.4}$$

Proof. Existence. We partly repeat the arguments in the proof of Theorem 4.1. For this, substituting (4.5) -(4.7) into (5.1), we have

$$ {}^cD^\alpha_{q,t}u_k(t) + \left(\lambda_k + m\right)u_k(t) = f_k, \tag{5.5}$$

with initial data

$$u_k(0) = \varphi_k, \tag{5.6}$$

and the final condition

$$u_k(T) = \rho_k, \tag{5.7}$$

for all $k \in I$.

By using (4.11) and (4.13) we find a general solution of the problem (5.5)-(5.7):

$$\begin{aligned} u_k(t) &\overset{(4.11)}{=} \varphi_k e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right) \\ &+ f_k \int_0^t t^{\alpha-1}\left(qs/t;q\right)_{\alpha-1}\varepsilon^{-q^{\alpha}s}e_{\alpha,\alpha}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)d_qs \\ &\overset{(4.13)}{=} \varphi_k e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right) \\ &+ \frac{f_k}{\lambda_k+m}\int_0^t D_{q,s}\left[\varepsilon^{-s}e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)\right]d_qs \\ &= \varphi_k e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)+\frac{f_k}{\lambda_k+m}\left[1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)\right], \end{aligned} \tag{5.8}$$

where the constant f_k needs to be determined.

Now, by using the conditions (5.6), (5.7) and (5.8) we get that

$$\begin{aligned} u_k(T) \overset{(5.7)}{=} \rho_k &\overset{(5.6)(5.8)}{=} \varphi_k e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right) \\ &+ \frac{f_k}{\lambda_k+m}\left[1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right)\right], \end{aligned}$$

so that

$$\begin{aligned} f_k &= \frac{\left(\lambda_k+m\right)\rho_k-\left(\lambda_k+m\right)\varphi_k e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right)}{1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right)} \\ &= \left(\lambda_k+m\right)\varphi_k-\frac{\left(\lambda_k+m\right)\varphi_k-\left(\lambda_k+m\right)\rho_k}{1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right)}. \end{aligned} \tag{5.9}$$

From (5.8)-(5.9) it is follows that

$$\begin{aligned} u_k &\overset{(5.8)(5.9)}{=} \varphi_k e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)+\varphi_k\left[1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)\right] \\ &- \frac{\left[\left(\lambda_k+m\right)\varphi_k-\left(\lambda_k+m\right)\rho_k\right]\left[1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)\right]}{\left(\lambda_k+m\right)\left[1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right)\right]} \\ &= \varphi_k+\frac{\left[\varphi_k-\rho_k\right]\left[e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)-1\right]}{\left[1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right)\right]}. \end{aligned} \tag{5.10}$$

Therefore, by using (4.5), (4.6), (5.9) and (5.10) we get that

$$u(t) \overset{(4.5)\text{-}(5.10)}{=} \varphi+\sum_{k\in I}\frac{\left[\varphi_k-\rho_k\right]\left[e_{\alpha,1}\left(-\left(\lambda_k+m\right)t^{\alpha};q\right)-1\right]\psi_k}{\left[1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right)\right]},$$

and

$$f \overset{(4.6)\text{-}(5.9)}{=} \mathcal{L}\varphi-\sum_{k\in I}\frac{\left(\lambda_k+m\right)\left[\varphi_k-\rho_k\right]\psi_k}{1-e_{\alpha,1}\left(-\left(\lambda_k+m\right)T^{\alpha};q\right)}.$$

Convergence. For $0 < t < T$, using the estimate (3.1), we get

$$\begin{aligned} 1 - e_{\alpha,1}\left(-\left(\lambda_k + m\right) T^{\alpha}; q\right) &\overset{(3.1)}{\geq} 1 - \frac{1}{1 + \Gamma_q\left(\alpha+1\right)^{-1}\left(\lambda_k + m\right) T^{\alpha}} \\ &= \frac{\Gamma_q\left(\alpha+1\right)^{-1}\left(\lambda_k + m\right) T^{\alpha}}{1 + \Gamma_q\left(\alpha+1\right)^{-1}\left(\lambda_k + m\right) T^{\alpha}}. \end{aligned}$$

Therefore

$$0 \leq 1 - e_{\alpha,1}\left(-\left(\lambda_k + m\right) t^{\alpha}; q\right) < 1, \quad 0 < t \leq T. \tag{5.11}$$

From (5.9), (5.10) and (5.11) we conclude that

$$|u_k| \overset{(5.10)(5.11)}{\lesssim} |\varphi_k| + |\rho_k|; \quad |f_k| \overset{(5.9)(5.11)}{\lesssim} \left(\lambda_k + 1\right)|\varphi_k| + \left(\lambda_k + 1\right)|\rho_k|. \tag{5.12}$$

We assume that $\varphi, \rho \in \mathcal{H}_{\mathcal{L}}^{d+2}$. Then using (4.2), (4.9) and (5.12) we derive that

$$\begin{aligned} \|f\|_{\mathcal{H}_{\mathcal{L}}^{d}}^2 &\overset{(4.2)}{=} \sum_{k\in I}\left|\left(1+\lambda_k\right)^{d/2}\left\langle f, \psi_k\right\rangle_H\right|^2 \\ &\overset{(5.12)}{\lesssim} \sum_{k\in I}\left|\left(1+\lambda_k\right)^{d/2+1}\left\langle \varphi, \psi_k\right\rangle_H\right|^2 + \sum_{k\in I}\left|\left(1+\lambda_k\right)^{d/2+1}\left\langle \rho, \psi_k\right\rangle_H\right|^2 \\ &\overset{(4.2)}{=} \|\varphi\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^2 + \|\rho\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^2 < \infty, \end{aligned} \tag{5.13}$$

and

$$\begin{aligned} \left\|\left(I+\mathcal{L}\right)u(t)\right\|_{\mathcal{H}_{\mathcal{L}}^{d}}^2 &\overset{(4.2)}{=} \sum_{k\in I}\left|\left(1+\lambda_k\right)^{d/2+1}\left\langle u(t), \psi_k\right\rangle_H\right|^2 \\ &\overset{(5.12)}{\lesssim} \sum_{k\in I}\left[\left|\left(1+\lambda_k\right)^{d/2+1}\left\langle \varphi, \psi_k\right\rangle_H\right|^2 + \left|\left(1+\lambda_k\right)^{d/2+1}\left\langle \rho, \psi_k\right\rangle_H\right|^2\right] \\ &\overset{(4.2)}{=} \|\varphi\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^2 + \|\rho\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^{d+2} < \infty. \end{aligned} \tag{5.14}$$

By using (4.9), (5.13) and (5.14) we find that

$$\begin{aligned} \|{}^{c}D_{q,t}^{\alpha} u(t)\|_{\mathcal{H}_{\mathcal{L}}^{d}}^2 &\overset{(4.9)}{\lesssim} \left\|\left(I+\mathcal{L}\right)u(t)\right\|_{\mathcal{H}_{\mathcal{L}}^{d}} + \|f\|_{\mathcal{H}_{\mathcal{L}}^{d}} \\ &\overset{(5.13)(5.14)}{\lesssim} \|\varphi\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^2 + \|\rho\|_{\mathcal{H}_{\mathcal{L}}^{d+2}}^{d+2} < \infty, \end{aligned}$$

which means that $u \in L_q^{\infty}\left([0,T]; \mathcal{H}_{\mathcal{L}}^{d+2}\right) \cap W_q^{\alpha}\left([0,T]; \mathcal{H}_{\mathcal{L}}^{d}\right)$ and $f \in \mathcal{H}_{\mathcal{L}}^{d}$, and the estimates (5.4) hold.

Uniqueness. The obtained solution (5.10)-(5.11) satisfies the equation (5.1) and the conditions (5.2)-(5.3).

We denote $v(t) := u_1(t) - u_2(t)$ and $f := f_1 - f_2$. Then $u(t)$ and f satisfy (5.1) and homogeneous conditions (5.2) and (5.3).

Applying the inner product on Hilbert space H and the equation (5.1) we obtain that

$$\begin{aligned}(1+\lambda_k)^c D^{\alpha}_{q,t} u_k(t) &= \langle {}^cD^{\alpha}_{q,t}\left[u(t)+\mathcal{L}u(t)\right], \psi_k\rangle_H \\ &= \langle -mu(t)+f, \psi_k\rangle_H \\ &= -mu_k(t)+f_k.\end{aligned}$$

Taking into account the homogeneous conditions (5.2) and (5.3), we find that $u_k(0)=0$ and $u_k(T)=0$. Therefore $f_k \equiv 0$ and $u_k \equiv 0$ for $k \in I$.

Further, by the completeness of the system $\{\psi_k\}_{k\in I}$ in H we obtain

$$f(t) \equiv 0, \quad u(t) \equiv 0,$$

for $t > 0$.

Hence, the uniqueness of the solution (5.10)-(5.11) is proved. The proof is complete. □

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